

01 Detect

Correlation baseline

DIR 1.74 · 4/5 rule fails

02 Discover

Causal structure search

6 algorithms · 4 families

03 Isolate

Direct vs. proxy pathway

Where the fix is decided

► DIRECT PATHWAY

Race → **Score**

Discriminatory bias embedded.

Intervention: revise internal logic.

► PROXY PATHWAY

Race → **Priors** → **Score**

Structural inequities inflate a “neutral” risk signal. **Intervention: upstream policy reform -removing Race will not eliminate this path.**

04 Quantify

The Causal Effect. Apply covariate adjustment to estimate the total and adjusted direct effect.

+1.64 attenuates to +0.53

EFFECT ATTENUATION UNDER CONTROLS

Total effect (raw)

+1.64

Partial adjustment (age, charge)

+1.14

Adjusted direct effect

+0.53

05 Validate

Bootstrap resampling

- ✓ 1,000 bootstrap draws
- ✓ CI excludes 0
- ✓ 95% CI [+0.40, +0.66] ·
- ✓ E-value 1.65

